

The Hand That Takes

*Hittite $d\bar{a}$ - as pre-PIE $*de$ + postposed $-a$, with iconographic evidence from Anatolian hieroglyph CAPERE*

Paper 4 of the *der-* project. Builds on Chavis & Claude (2026a, 2026b, 2026c).
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Abstract

The first three papers of the *der-* project reconstructed pre-PIE $*de$ - "hand, open palm" and decomposed both the body-part noun $*der$ - and the give-verb $*deh_3$ - into the same bare hand-element extended by different morphemes (Papers 1–3). The present paper carries the framework into Anatolian. We argue that Hittite $d\bar{a}$ - "to take" preserves a fossilized univerbation of pre-PIE $*de$ - with a postposed directional element $*a$ "toward (the speaker)," contracted by hiatus to long $*\bar{a}$, at a stage before the verbal extension $*-h_3$ - had been added. The take/give polarity thereby resolves into a structural opposition: bare $*de$ - "extend (the hand)" yields the giving direction in post-Anatolian PIE, while $*de$ - extended by directional $*a$ "toward the speaker" yields the taking direction preserved in Hittite $d\bar{a}$ -.

The argument rests on four convergent strands. First, synchronic Hittite already has a productive system of directional reversal on the bare da - stem: the indissociable preverbs u - "hither" and pe - "thither" pair uda - "to bring here" against $p\bar{e}da$ - "to send away" against the simplex $d\bar{a}$ - "to take" (Kloekhorst 2008: 928–930; Hoffner & Melchert 2008: §20.7). Second, Hittite has a productive paradigmatic opposition between Set 1 directional adverbs in $-a$ ($anda$ "into," $appa$ "back," $katta$ "down," $par\bar{a}$ "forth," $\bar{s}ar\bar{a}$ "up") and Set 2 locative adverbs in $-an$ ($andan$ "within," $appan$ "behind," $kattan$ "below"); the directional $-a$ is the same morpheme as the Hittite allative case ending. Third, Indo-European in its earliest reconstructable strata is postpositional and OV, with Anatolian and Tocharian directly preserving postpositional inheritance. Fourth, Norbruis (2021: 243–249) argues independently that the original meaning of $*deh_3$ - was "to take" and that "to give" is a post-Anatolian semantic innovation; his chronology aligns directly with the morphosyntactic chronology proposed here.

Independent corroboration comes from the indigenous Anatolian writing system. Hieroglyph L 41 is iconographically a grasping hand and serves both as the logogram CAPERE "to take" and as a syllabogram whose phonetic value Yakubovich (2008) derives acrophonically from Hittite $d\bar{a}$ - "to take." Under the present framework, where $d\bar{a}$ - derives from pre-PIE $*de$ - (hand) + postposed $*a$, the hand iconography is not a coincidence of universal cognition but an

iconographic preservation of the verb's etymological source. Bauhaus's (2019) reconstruction of a paradigmatic opposition between locative **r* and directive **o* on the same nominal roots — **h₁upér / *h₁upó, *k(e)nter / *knto* and related pairs — provides published external precedent for the structural opposition the framework reconstructs across Papers 3 and 4: locative **de-r-* (the body-part noun) versus directive **de + a* (this paper's Anatolian take-verb). The framework is offered for examination, not as a refutation of the standard reconstruction.

1. Introduction

This is the fourth paper in the *der-* project. The first three established that pre-PIE had a body-part etymon meaning "hand, open palm," that this etymon underlies the cognate cluster Greek δῶρον/Hesychian δάρις/Celtic **durno-/Latvian *dūre* (Paper 1), that the same etymon underlies the PIE contrastive suffix **-tero-* via grammaticalization through HAND > SIDE > BINARY-SPATIAL-CONTRASTIVE (Paper 2), and that the body-part noun **der-* decomposes compositionally into the bare hand-element **de-* plus a locative-spatial extension **r-* while the give-verb **deh₃-* decomposes in parallel as the same bare **de-* plus a verbal extension **-h₃-* (Paper 3).

The framework so far has remained within the post-Anatolian PIE that the standard handbooks reconstruct. The present paper carries it one step deeper, into Anatolian itself. Two facts about Anatolian, taken together, demand engagement. The first is well known: Hittite *dā-* / cognate Luwian *lā-* reflects the same etymon as the give-verbs of the rest of the family but means "to take," not "to give." The second is less often emphasized: Hittite preserves a productive synchronic system in which the bare *da-* stem combines with directional preverbs to specify deictic orientation, with *u-* "hither" yielding *uda-* "to bring here" and *pe-* "thither" yielding *pēda-* "to send away." The construction-type the framework has been hypothesizing — directional reversal of the bare hand-verb by an attached element — is therefore attested in synchronic Hittite on the same root.

The thesis of this paper is that the bare *dā-* "to take" is itself the residue of a similar but earlier construction, in which a postposed directional element **a* "toward the speaker" attached to the bare hand-element **de-* and underwent univerbation by hiatus contraction **e + a > *ā* in pre-Anatolian. The verbal extension **-h₃-* of Paper 3 is, on this view, a post-Anatolian innovation: a derivational suffix added to the bare hand-element **de-* to create a verbal stem **deh₃-* that yielded the give-verbs of the other branches. Anatolian split off before that extension was added.

The proposal is supported by independent evidence streams: the synchronic Hittite *u-/pe-* prefix system on the bare *da-* stem (§3); the productive Hittite Set 1 / Set 2 directional-locative contrast (§4); the postpositional and OV character of

earliest PIE (§5); Bauhaus's (2019) independently published reconstruction of locative-directive paradigmatic pairs on the same nominal roots (§5a); Norbruis's (2021) independent semantic chronology (§7); the iconographic evidence from Anatolian hieroglyph L 41 (§8a); and Boley (2007) as historiographic precursor (§8).

The proposal does not require Kloekhorst's (2013) glottalization analysis of *dā-* to be wrong. The two analyses are observationally close in attested Anatolian. We treat the Kloekhorst-Norbruis analysis as one viable interpretation and our proposal as a second, identifying where they diverge in their predictions and where they converge (§9).

2. The standard reconstruction of Hittite *dā-*

Kloekhorst's (2008: 928–930) entry on Hittite *dā-/d-* "to take" gives the canonical Old Hittite paradigm. The paradigm shows stem alternation between long-*ā* forms in the singular and short-vowel forms in the plural.

Form	Hittite	Notes
1sg pres.	<i>dāḥḥe / dāḥḥi</i>	strong stem <i>dā-</i>
2sg pres.	<i>dātti</i>	strong stem <i>dā-</i>
3sg pres.	<i>dāi</i>	strong stem <i>dā-</i>
1pl pres.	<i>tumēni, dāweni</i>	weak stem, /u/ or /aw/ vocalism
2pl pres.	<i>dattēni</i>	weak stem
3pl pres.	<i>danzi</i>	weak stem, /a/ vocalism

Table 1. Old Hittite paradigm of *dā-* "to take" (after Kloekhorst 2008: 928–930).

Kloekhorst derives this paradigm from PIE **deh₃-* with regular ablaut: the strong stem **dóh₃-* gives long *ā* via the regular Anatolian outcome of **ō* < **oh₃*; the zero grade **dh₃-* gives /a/ via regular laryngeal vocalization in *danzi* < **dh₃-énti*; the /u/ in *tumēni* comes from regular vocalization of zero-grade **dh₃-uén*i with **u*-coloring.

One observation from Kloekhorst is directly relevant to the present paper. The compound verbs *uda-* "to bring here" and *pēda-* "to bring away," formed with the indissociable prefixes *u-* and *pe-*, preserve a more archaic plural paradigm than the simplex. He notes (2008: 930): "the derivatives *peda-/ped-* 'to bring (away)' and *uda-/ud-* 'to bring (here)' show forms that are more archaic, e.g. *petumen* and *utumen* vs. *dāwen*, *petišten* vs. *dāttēn*, *peter* and *uter* vs. *dāer*." The paradigm of *dā-* is therefore demonstrably innovative in its plural; the older state, accessible through the compounds, had a distinct stem alternation. This archaic plural will be relevant in §6.

Melchert (1994: 89) reconstructs Proto-Anatolian **dō-* for the take-verb and identifies the long **ō* as the regular outcome that becomes Hittite long *ā*. His section on laryngeal-in-coda treatment (§4.1.6.3.1) documents that voiceless **h* becomes voiced in word-final position and is then lost with compensatory lengthening: PIE **-eh₂#* > **-aH#* > Anatolian **-ā*. The principle that long vowels in Anatolian can derive from contracted vowel-plus-laryngeal or vowel-plus-vowel sequences is therefore well established for the family.

3. The Hittite *u-/pe-* prefix system

Hoffner and Melchert (2008: §20.7) describe the Hittite *u-/pe-* preverb system synchronically: "In Hittite only two truly inseparable preverbs exist: *u-* 'here' and *pe-* 'there', attested in the pairs: *uda-* 'to carry here' and *pēda-* 'to carry away'; *uiya-* 'to send here' and *peya-* 'to send away'; *unna-* 'to drive here' and *penna-* 'to drive away'; *uwate-* 'to lead here' and *pehute-* 'to lead away'."

Within the take-verb paradigm specifically, the system gives: bare *dā-* "to take" (long *ā*, deictically unspecified in attested Hittite); *uda-* "to bring here" = *u-* "hither" + *da-* (short vowel); *pēda-* "to send away" = *pe-* "thither" + *da-* (short vowel). Two features of this synchronic system bear on the framework.

First, the construction-type the framework hypothesizes — directional reversal of the take-verb by an attached element — is real and productive in attested Anatolian. The objection that pre-PIE **de-* + directional **a* is morphosyntactically unattested loses force when the same kind of directional-reversal construction is documented for the same root in attested Hittite.

Second, the long *ā* of bare *dā-* contrasts with the short vowel of the prefixed forms *uda-* and *pēda-*. When the directional element is overt, the verbal stem has short vowel; when the verb stands alone, the vowel is long. Under the standard reconstruction this contrast follows from o-grade vs. zero-grade ablaut. Under the present proposal it falls out of a different mechanism: the prefixed compounds preserve the bare stem attached to a prefixed directional, while the simplex preserves the fossilized univerbation **de + a* > **dā* with the long vowel as the trace of the contracted directional. The two analyses are observationally equivalent for the synchronic Hittite forms but make different claims about the historical derivation.

4. The Hittite *-a / -an* opposition: directional versus locative

Hoffner and Melchert (2008: §20.3) describe a productive Hittite paradigmatic contrast between two paired sets of local adverbs. Set 1 ("Place to Which"): *anda* "into," *āppa* "back," *katta* "down," *parā* "out, forth," *šarā* "upward." Set 2 ("Place Where"): *andan* "within, inside," *āppan* "behind, after," *kattan* "below," *pēran* "in front," *šer* "above." Set 1 ends in *-a*; Set 2 ends in *-an*. Set 1 marks direction-toward; Set 2 marks location-at. The contrast is paradigmatic: each Set 2

adverb has a Set 1 counterpart, with the formal alternation *-a* : *-an* mapping consistently onto the directional/locative semantic contrast.

The *-a* suffix marking direction-toward is the same morpheme as the Hittite allative case ending *-a* attested productively in nominal forms: *aruna* "to the sea" (allative) vs. *aruni* "at the sea" (locative); *nepisa* "to heaven" vs. *nepisi* "in heaven." Across both the local-adverb system and the nominal case system, *-a* marks the direction-toward semantics that the present paper proposes for the directional element underlying Hittite *dā-*.

The deeper PIE source of this Hittite directional *-a* is contested. Melchert (2017) reconstructs the Anatolian allative as PIE **eh₂*, basing the argument on Greek *χαμαί* "to the ground" and Lycian infinitives in *-na*. Paper 3 (§2.3) addresses the *χαμαί* question and argues that the locative reading is preferable, which would leave space for a **de--*derived allative element. What matters here is that Anatolian has a productive directional morpheme *-a* whatever its deeper source, and the present paper proposes that the long *ā* of bare *dā-* "take" is a fossilized univerbation involving exactly this morpheme.

5. PIE postpositional syntax: the early-stratum context

The framework's proposal requires that pre-PIE allowed postposed adverbial directional elements to appear after verbal stems. This is not an exotic configuration; it is the unmarked one for the earliest reconstructable layers of Indo-European syntax.

Lehmann's (1974) *Proto-Indo-European Syntax* argued on typological grounds that PIE was an OV language whose unmarked phrasal pattern required postposed adpositions and modifiers. Friedrich (1975) refined the typology. The unmarkedness of postposed adpositions in the earliest stratum has held up across subsequent critique (Hock 1996; Kiparsky 1995). What is at stake for the present paper is not strict OV reconstruction but the simpler claim that postposed adverbial elements were available in earliest reconstructable PIE.

Four direct attestations support this. Hittite is uniformly postpositional in its adpositional system: all Hittite adpositions are postpositions in nominal phrases (*parni anda* "into the house," with *anda* after the dative-locative *parni*). As the earliest-attested Indo-European branch (Kloekhorst & Pronk 2019), Anatolian's postpositional pattern is a strong indicator of the PIE state. Tocharian, the second-earliest branch to split off under most current chronologies, is also predominantly postpositional; its complete reorganization into postpositional paradigms cannot be derived from a strictly prepositional ancestor without ad hoc machinery. Vedic Sanskrit preserves substantial postpositional flexibility: the Rigvedic adverbial-particle system places adverbs in flexible positions, and the shift to fixed prepositional usage is a Classical Sanskrit development. Latin

preserves postpositional archaisms in fossilized form: *mēcum* "with me," *tēcum* "with you," *nhbiscum* and *vnhbiscum* all have *cum* postposed to the personal pronoun. Legal-formulaic postposed *dē* in expressions like *quā dē causā* preserves the same archaism.

Homeric Greek shows the relevant change in real time. In Homer the so-called "preverbs" remain separable from the verb (the famous tmesis: κατὰ δάκρυα λείβων "shedding tears down," with κατὰ free-standing rather than fused into Classical καταλείβων). By Classical Greek these elements have unverbated with the verb stem. The trajectory is clear: free adverbial particles > fixed preverbs > unverbated prefix-plus-verb compounds. The earlier state, even within attested Greek, is one of free adverbial elements in flexible positions. This is the state the framework proposes for pre-PIE.

Pre-PIE in its earliest reconstructable state therefore had free adverbial directional elements that could appear in postposed positions, with subsequent fossilization as bound prefixes or suffixes through unverbation. The construction we propose for the bare hand-element — **de-* + postposed directional **a* > fossilized **dā-* — is a special case of this documented PIE pathway. It is not asking for a novel suffix slot. It is asking for an originally free postposed adverbial to fossilize with a preceding verbal stem and contract by hiatus — the same kind of process by which Latin *mēcum* arose, proposed here at an earlier date.

5a. Bauhaus (2019): external precedent for the structural pair

Paper 3's new §3a documents Bauhaus's (2019) reconstruction of a paradigmatic opposition between locative **-r* and directive **-o* on the same nominal roots in early PIE. The pairs include **h₁upér* "on top" vs. **h₁upó* "to the top," **k(e)nter* "at the underside" vs. **knto* "downwards," **ndher* "at the bottom" vs. **ndho* "to the bottom," and **h₂eper* "in the back" vs. **h₂epo* "to the back." Bauhaus summarizes: "all this strongly suggests that we are dealing with a paradigmatic opposition ... whereas *-o* in its allative function could keep its productivity at least until Anatolian, *-r* has apparently ceased to be productive in all attested languages" (Bauhaus 2019: §5).

This structural observation is directly relevant to the present paper. The framework's proposal across Papers 3 and 4 reconstructs exactly this kind of paradigmatic opposition on the bare nominal root **de-* "hand": locative **de-r-* (the body-part noun, Paper 3) versus directive **de + a* (the Anatolian take-verb, this paper). The framework is not proposing a one-off peculiarity of the hand-root. It is applying a documented structural pattern of PIE morphology — Bauhaus's locative/directive pair — to a specific lexical item. The published reconstruction covers the type; the framework adds the instance.

Bauhaus's chronological observation strengthens the framework's Anatolian argument specifically. He notes that the directive **-o* "could keep its productivity at least until Anatolian," while the locative **-r* "ceased to be productive in all attested languages." Under the framework's architecture, this is exactly the observed distribution: the locative member **de-r-* was fossilized as a body-part noun before the Anatolian split (it survives as a lexical item in Greek, Celtic, and Baltic); the directive member **de + a* remained productive into Anatolian, where it contracted to **dā-* and was preserved as an active verb. Bauhaus's typological generalization predicts this asymmetry; the framework's Anatolian evidence instantiates it.

6. The proposal: pre-PIE **de-* + postposed **a* > Hittite *dā-*

On the framework of Papers 1–3, pre-PIE had a bare hand-element **de-* that functioned both nominally (extending in **der-* with locative **-r-*) and verbally (extending in post-Anatolian **deh₃-* with the verbal **-h₃-*). The bare element appears independently as the directional particle **de/do* "to, toward" across the family — preserved as Greek allative *-δε*, Old Church Slavonic *do*, English *to*, and as the Hittite directional-toward suffix in Set 1 of the local-adverb system (Paper 3, §2).

In its earliest reconstructable stratum, pre-PIE was postpositional and OV (§5). At that stage, the bare hand-element **de-* could appear as a verbal stem, and a directional adverbial element **a* could appear postposed to it to specify deictic orientation:

****de-* + **a* → **de a* → **dā* (by hiatus contraction **ea* > **ā*)**

The semantics of the construction are: bare **de-* "extend (the hand)" in the unmarked direction (= give, away from the speaker) plus directional **a* "toward the speaker" yields the deictic-reversed meaning "extend toward oneself" = "take, receive." The construction is parallel to the synchronic Hittite *uda-* "to bring here" = bare verbal stem + directional "hither" = "to take in the speaker's direction," but with the directional element in postposed rather than prefixed position, reflecting the older Anatolian morphosyntactic state.

The contraction **e + a* > **ā* is consistent with the well-attested Anatolian phonological treatment of vowel sequences. Melchert (1994: §4.1.6.3.1) documents that word-final **-eh₂* was treated as voiced **-aH* after laryngeal voicing in word-final position and then lost with compensatory lengthening, yielding **-ā*. The principle that long vowels in Anatolian can derive from contracted vowel-plus-laryngeal or vowel-plus-vowel sequences is well established. The specific contraction we propose is consistent with this general pattern and does not require a new sound rule.

The proposal makes a specific prediction about the plural forms of the paradigm. The plural forms with /u/-vocalism (*tumēni*) and /a/-vocalism (*danzi*) are the older fossilized forms and reflect the bare verbal stem **d-* without the directional **a*, attached to inflectional endings whose vowels produced the surface /u/ and /a/. The long-*ā* of the singular subsequently spread into the plural in late Old Hittite (yielding *dāwen*, *dāer*) — exactly the innovation Kloekhorst documents, and equally well-predicted by the present analysis as by the standard one.

7. Norbruis (2021): independent semantic support

Norbruis (2021: 243–249), in his Leiden dissertation chapter on the etymology of **deh₃-*, argues independently for a chronology closely aligned with the morphosyntactic chronology proposed here. His conclusion: "In this chapter it is argued that the verb **deh₃-* did not mean 'to give' in PIE, but 'to take', as in Anatolian. Although the verb did not survive in any evident way in Tocharian and Germanic, the semantic shift to 'to give' is likely to be an innovation of post-Anatolian IE."

Norbruis works through every parallel adduced for a give > take semantic development and shows they actually support the opposite direction. He proposes that the original meaning was "to take," with the post-Anatolian shift to "to give" running through an intermediate stage "to bring/deliver," paralleled by English *take to*, Biblical Hebrew *lāqah*, and Old Norse *fá* "to take; to give." His reconstruction of the semantic development runs: "to take" > (with Goal or Beneficiary constituent) "to bring, convey, deliver" > (when the constituent was animate) "to hand, to give." The Anatolian branch preserves only the original meaning because it split off before the development started.

Norbruis's chronology aligns directly with the framework proposed here. Anatolian split off not only before the semantic shift but also, on the framework's account, before the **-h₃-* verbal extension was added. Where Norbruis stays with the standard formal reconstruction (PA **doH-*, with the laryngeal preserved as glottalization), the framework proposes that the long *ā* of *dā-* reflects fossilized **de + a* rather than **dóh₃-*. The two analyses are decoupleable: Norbruis's semantic argument is fully independent of his formal reconstruction, and the framework's formal proposal can stand or fall independently of the semantic chronology. We take the coincidence of formal and semantic chronology as mutually reinforcing.

8. Boley (2007): historiographic precursor

The proposal that **deh₃-* derives from a directional particle is not unprecedented. Boley (2007), in a paper on the Anatolian *dā-* / *pai-* "give" complex, previously proposed deriving the give/take root from "a directionally ambivalent particle reconstructed on the basis of Latin *dē* 'from' and Greek *-δε* 'to'" (as reported by

Norbruis 2021: 245 n. 9). The same Greek allative clitic $-\delta\epsilon$ that Paper 3 adduced for **de-* thus appears in Boley's earlier proposal as well.

Norbruis (2021: 245 n. 9) dismisses Boley's specific arguments as unsound, and we agree with much of his criticism. The notion of a "directionally ambivalent particle" with both "from" and "to" senses simultaneously is not well motivated. What survives the criticism, and what the present paper picks up, is the structural insight: the give/take verb of PIE derives from a directional particle. Boley got the connection itself right even though her specific machinery was unsatisfactory.

The present paper offers a more constrained version of Boley's intuition. Rather than a single "directionally ambivalent" particle, we propose a directional-toward element **a* which combined with the bare hand-element **de-* to yield the take-meaning, while the bare element alone in post-Anatolian developed the give-meaning through the **h₃-* verbal extension. The directional contrast is specific, the morphology is more precisely specified, and the chronology is supplied by Norbruis (2021). We credit Boley as the historiographic precursor to the directional-particle hypothesis while departing from the specific mechanisms she proposed.

8a. Iconographic evidence: Anatolian hieroglyph CAPERE

An independent line of evidence comes from the indigenous Anatolian writing system. Anatolian hieroglyphic sign L 41 (Laroche numeration) is iconographically a "taking" or "grasping" hand. It functions both as the logogram CAPERE "to take" — used to write words for taking, most commonly the Luwian verb *lā-* — and as a syllabogram with phonetic value /tā/ (Rieken 2008; Yakubovich 2008) or [ḏa] in some recent analyses. The Anatolian hieroglyphic script is indigenous to central Anatolia, with no demonstrable connection to Hittite cuneiform and no graphical derivation from Egyptian hieroglyphs (Hawkins 2000; Payne 2015). The iconographic choice — a grasping hand for the verb "to take" — is therefore a product of Anatolian, specifically Hittite-Luwian bilingual, cognitive associations.

The phonetic value of L 41 was acrophonically derived: script-creators assigned the sign a syllabic value matching the initial syllable of a verb meaning "to take." The senior published analysis is Yakubovich (2008), who derives the value /tā/ from Hittite *dā-* "to take" in the bilingual Hittite-Luwian environment of script development at the court of Ḫattuša. Under the present framework — where Hittite *dā-* derives etymologically from pre-PIE **de-* (the bare hand-element) extended by postposed directional **a* — the hand iconography of L 41 is not a coincidence of universal cognition but an iconographic preservation of the verb's etymological source. The Anatolian script-creators chose a hand to depict *dā-* because the verb was, in its deep history, a hand-derivation.

A competing analysis is offered by Kloekhorst (2019), who derives the value [ð̥a] of L 41 from the Luwian verb *lā-* "to take," reconstructed as Proto-Anatolian **dh₃-* with the laryngeal **h₃* preserved as fricativization on the dental before [ð̥] developed to /l/ word-initially in Luwian. This re-analysis depends on the word-initial laryngeal-trace mechanism that the framework engages in §9. As discussed there, the underlying mechanism is contested by Melchert (2015) and by Sideltsev & Molina (2021). The Kloekhorst-Norbruis re-analysis of L 41 depends on contested mechanisms; Yakubovich's senior acrophonic analysis does not. We treat Yakubovich's analysis as the cleaner basis for interpreting the iconographic evidence.

What the iconographic evidence does and does not establish should be stated precisely. It does establish that Anatolian speakers, when designing their indigenous script, cognitively associated the verb "to take" with hand iconography; at the level of representational choice, the hand-take connection is unambiguous. It does establish that on the senior acrophonic analysis the sign's phonetic value derives from precisely the verb form the framework predicts. It does not, by itself, decide between the framework and the standard reconstruction at the level of historical phonology. The iconographic evidence is corroborative — strong corroboration on the senior analysis, weaker on the contested re-analysis — not decisive.

9. Reconciliation with the standard reconstruction

The standard reconstruction of Hittite *dā-* — Kloekhorst (2008, 2013), Norbruis (2021) — derives the verb from PA **doH-* / **dH-* with the laryngeal **h₃* preserved as glottalization of the initial stop. The present paper proposes an alternative derivation from pre-PIE **de + a* via hiatus contraction, with the **-h₃-* extension absent from the Anatolian verb because it was a post-Anatolian innovation. The two analyses are observationally close in attested Anatolian but diverge in three theoretical commitments.

They diverge on the *source of the long ā*: the standard analysis derives it from o-grade **dóh₃-* via laryngeal lengthening; the present analysis derives it from **de + a* via hiatus contraction. Both mechanisms are documented in Anatolian phonology. They diverge on the *status of *h₃*: the standard analysis treats it as a real laryngeal phoneme present in Anatolian, surfacing as glottalization on the initial stop and as the Luwian *l-* of *lā-*; the present analysis treats it as a post-Anatolian derivational suffix absent from the Anatolian verb. They diverge on the *directional reversal mechanism*: the standard analysis explains the take/give polarity as a post-Anatolian semantic shift on a single root with no morphological correlate; the present analysis explains the polarity as a structural opposition between bare **de-* (yielding give in post-Anatolian) and **de + a* (yielding take in Anatolian), with a morphological correlate that aligns with the synchronic Hittite

u-/pe- / simplex three-way contrast.

The standard analysis has the advantage of formal economy within the laryngeal-theoretic framework and derives Hittite *dā-* / Luwian *lā-* from a single PIE root with established sound laws. It has the disadvantage that the take/give polarity is not given a structural source — it is a semantic accident of post-Anatolian PIE rather than a structural feature of the etymology — and that Kloekhorst's (2013) glottalization analysis is contested within the field.

Three considerations weigh against the glottalization analysis specifically. First, Melchert (2015) argues explicitly: "There is no evidence for word-initial **h₁* preserved as [ʔ] in Hittite and Luvian, contra Kloekhorst (2004, 2006a: 77–81 and 2008: 205 and passim) and Simon (2010 and 2013)." Second, Sideltsev & Molina (2021) raise independent methodological objections to Kloekhorst's broader prosodic-phonological framework — two independent scholarly traditions (the American Anatolianist school at UCLA and the Russian Anatolianist school in Moscow) converging on skepticism. Third, the proposed sound change **dʰ-* > Luwian /l-/ has no independent parallels elsewhere in Anatolian phonology; it is an ad hoc derivation invented to handle this single etymology.

Additionally, **h₃* has the weakest independent evidence base of the three PIE laryngeals. **h₁* leaves demonstrable *e*-coloring effects across the family; **h₂* leaves clear *a*-coloring and compensatory-lengthening patterns; **h₃* reduces primarily to *o*-coloring in Greek (which some analysts derive differently) and the contested Anatolian glottalization argument. Kloekhorst's argument stands at the most vulnerable point of laryngeal theory. None of these observations refutes the standard analysis; together they shift the burden of proof toward establishing the glottalization-as-laryngeal-trace interpretation against alternatives that do not require positing a laryngeal in the Anatolian verb at all.

Under the present account, **h₃-* was not present in the Anatolian take-verb at any stage. Whatever phonetic feature Kloekhorst (2013) detects on Old Hittite *dā-* must therefore have a different historical source: scribal practice in Old Hittite cuneiform sign-selection, whose phonological interpretation is methodologically variable; a feature of the *hi*-conjugation class to which *dā-* belongs (Jasanoff 2003 derives *hi*-conjugation verbs from PIE perfect stems, which would supply distinctive morphology independent of laryngeal preservation); or some combination of these. The interpretation of the observation, not the observation itself, is what divides the two analyses.

We do not claim that the present analysis is decisively superior. What we claim is that it is structurally motivated, evidentially grounded, consistent with the published Indo-Anatolian chronology (Norbruis 2021) and the scholarly precedent (Boley 2007), and provides a structural source for the take/give polarity that the standard account handles only semantically. It is offered as a viable alternative,

not a refutation.

10. Probability estimate

Following the practice of Papers 1–3, we close with an explicit conditional probability assessment.

Conditional on Papers 1, 2, and 3 being correct — that pre-PIE had a body-part etymon **der-* "hand" (Paper 1), that **-tero-* descends from this etymon via HAND > SIDE > BINARY-SPATIAL-CONTRASTIVE (Paper 2), and that **der-* decomposes into bare **de-* plus **-r-* while **deh₃-* decomposes into bare **de-* plus **-h₃-* (Paper 3) — our estimate of the probability that the substantive thesis of this paper is correct in its basic outline is approximately **38–48%**. The conditional probability is higher than for Paper 3 because the postpositional argument removes the central morphosyntactic objection, the Hittite *u-/pe-* system provides in-language attestation of directional reversal on the bare *da-* stem, the iconographic CAPERE evidence provides independent corroboration, and Bauhaus's (2019) published locative-directive paradigmatic pairs provide external precedent for the structural opposition the framework reconstructs across Papers 3 and 4.

The marginal probability — unconditioned, integrating over the probability that Papers 1, 2, and 3 are also correct — is approximately the product: 25–35% (Paper 1 v26) × 35–45% (Paper 2 conditional on Paper 1) × 32–42% (Paper 3 v4 conditional on Papers 1 and 2) × 38–48% (Paper 4 conditional on Papers 1–3) ≈ 1.1–2.6%. The marginal figure is in the low single digits and reflects the cumulative ambition of the framework. Each successive layer of pre-PIE morphosyntactic reconstruction lowers the marginal probability. We continue to regard the central claim as more likely wrong than right at the marginal level, and to regard it as interesting enough, and the evidential structure dense enough, that publication for examination by other scholars is warranted.

11. Authorship and contributions

This is the fourth paper in the *der-* project. It builds on Papers 1, 2, and 3 (Chavis & Claude 2026a, 2026b, 2026c) and continues the same co-authorship arrangement.

Chavis's contributions. The substantive intellectual direction of Paper 4 originates with Chavis. The hypothesis that Hittite *dā-* "to take" preserves an earlier construction in which the bare hand-element **de-* was extended by a postposed directional element is Chavis's. The decision to bring the framework into Anatolian, to address the take/give polarity through morphosyntactic rather than purely semantic mechanisms, and to recognize PIE postpositional syntax as removing the central objection to the proposal are all Chavis's. The structure of the paper, the framing of the relationship to Kloekhorst and Norbruis, the decision to include Boley (2007) as historiographic precursor, and the placement of Paper

4 as the closing paper of the four-paper project reflect Chavis’s overall vision for the project.

Claude’s contributions. Claude’s contributions are primarily in source synthesis, exposition, and evaluation. Claude identified and surfaced the relevant published literature: Kloekhorst (2008) on the Hittite paradigm; Hoffner & Melchert (2008) on the synchronic Hittite preverb and adverb systems; Melchert (1994) on Anatolian phonological rules; Norbruis (2021) on the semantic chronology and the Indo-Anatolian hypothesis; Boley (2007) as historiographic precursor (via Norbruis 2021: 245 n. 9); Lehmann (1974) and Friedrich (1975) on PIE postpositional syntax; Yakubovich (2008) on the acrophonic source of CAPERE; Bauhaus (2019) on the locative-directive paradigmatic pairs providing structural precedent for the framework’s opposition across Papers 3 and 4. Claude drafted the prose from Chavis’s structural and substantive direction and is responsible for the specific framings of arguments and local rhetorical choices. The probability estimate in §10 is Claude’s own.

Claude is an AI system without persistent memory between sessions; the intellectual continuity of the project is supplied by Chavis through systematic transcript management. The collaboration is genuine. The argument was built together. Its strengths and weaknesses belong to both authors, with the responsibility resting with Chavis.

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